Networking For Hackers

Complete Networking Guide for Ethical Hacking by Cyber Mind Space

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1. Networking Fundamentals

What is Networking?

Networking is the practice of connecting multiple devices (computers, servers, switches, routers) to share resources, data, and services. In cybersecurity, understanding networking is crucial because:

- **Attack Surface**: Every network connection is a potential entry point for attackers
- **Data Flow**: Understanding how data moves helps identify vulnerabilities
- **Defense Strategies**: Proper network design prevents many attacks
- **Incident Response**: Network analysis helps in forensic investigations

OSI Model vs TCP/IP Model

OSI Model (Open Systems Interconnection)

Mnemonic: "Please Do Not Throw Sausage Pizza Away"

Layer	Name	Function	Examples
7	Application	User interface, network services	HTTP, HTTPS, FTP, SSH, DNS
6	Presentation	Data formatting, encryption, compression	SSL/TLS, JPEG, GIF
5	Session	Establishes, manages sessions	NetBIOS, RPC, SQL sessions
4	Transport	End-to-end delivery, error recovery	TCP, UDP
3	Network	Routing, logical addressing	IP, ICMP, ARP, OSPF
2	Data Link	Frame formatting, error detection	Ethernet, PPP, Switch operations
1	Physical	Bits, electrical signals	Cables, hubs, repeaters

TCP/IP Model (Internet Protocol Suite)

Layer	Name	OSI Equivalent	Function
4	Application	5,6,7	Application services
3	Transport	4	End-to-end communication
2	Internet	3	Routing and addressing
1	Network Access	1,2	Physical network access

Types of Networks

By Geographic Scope:

- **PAN (Personal Area Network)**: 1-10 meters (Bluetooth, USB)
- **LAN (Local Area Network)**: Single building/campus
- **MAN (Metropolitan Area Network)**: City-wide
- **WAN (Wide Area Network)**: Country/continent-wide
- **Internet**: Global network of networks

By Ownership:

- **Private Networks**: Controlled by single organization

- **Public Networks**: Open to general public (Internet)
- **Hybrid Networks**: Combination of private and public

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2. Network Devices

Core Network Devices

Hub (Legacy Device)

- **Function**: Repeats data to all ports (broadcast domain)
- **OSI Layer**: Physical (Layer 1)
- **Security Risk**: All devices receive all traffic (easy eavesdropping)
- **Status**: Mostly obsolete, replaced by switches

Switch

- **Function**: Forwards frames based on MAC addresses
- **OSI Layer**: Data Link (Layer 2)
- **Key Features**:

    - Maintains MAC address table
    - Creates separate collision domains
    - Supports VLANs
- **Security Benefits**: Reduces broadcast traffic, enables port security

Router

- **Function**: Routes packets between different networks
- **OSI Layer**: Network (Layer 3)
- **Key Features**:
    - Uses routing tables
    - Connects different network segments
    - NAT (Network Address Translation)
    - Access Control Lists (ACLs)

Firewall

- **Function**: Controls network traffic based on security rules
- **Types**:

    - **Stateless**: Examines each packet independently
    - **Stateful**: Tracks connection states
    - **Next-Gen**: Application-layer inspection, intrusion prevention
- **Placement**: Network perimeter, internal segmentation

IDS (Intrusion Detection System)

- **Function**: Monitors and alerts on suspicious activities
- **Types**:
    - **Network-based (NIDS)**: Monitors network traffic
    - **Host-based (HIDS)**: Monitors individual systems
- **Detection Methods**: Signature-based, anomaly-based, behavioral

IPS (Intrusion Prevention System)

- **Function**: IDS + active blocking of threats
- **Advantage**: Real-time threat mitigation
- **Disadvantage**: Can block legitimate traffic (false positives)

Proxy Server

- **Function**: Intermediary between clients and servers
- **Types**:
  - **Forward Proxy**: Clients → Proxy → Internet
  - **Reverse Proxy**: Internet → Proxy → Internal servers
- **Benefits**: Caching, content filtering, anonymity

Load Balancer

- **Function**: Distributes incoming requests across multiple servers
- **Methods**: Round-robin, weighted, least connections
- **Benefits**: High availability, performance optimization

Network Device Diagram

```
Internet
    |
[Firewall] ↔ [IDS/IPS]
```

```
    |
[Router]
    |
[Core Switch]
    |
├─[Access Switch]─[Computers]
├─[Access Switch]─[Servers]
└─[Wireless AP]─[Mobile Devices]
```

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3. IP Addressing

IPv4 Addressing

IPv4 Structure

- **Format**: 32-bit address divided into 4 octets
- **Example**: 192.168.1.100
- **Total Addresses**: ~4.3 billion (2^{32})

IPv4 Classes

Class	Range	Default Subnet Mask	Network Bits	Host Bits	Max Networks	Max Hosts/Network
A	1-126	255.0.0.0 (/8)	8	24	126	16,777,214
B	128-191	255.255.0.0 (/16)	16	16	16,384	65,534
C	192-223	255.255.255.0 (/24)	24	8	2,097,152	254
D	224-239	Multicast	-	-	-	-
E	240-255	Reserved	-	-	-	-

Private IP Ranges (RFC 1918)

- **Class A**: 10.0.0.0/8 (10.0.0.0 - 10.255.255.255)
- **Class B**: 172.16.0.0/12 (172.16.0.0 - 172.31.255.255)
- **Class C**: 192.168.0.0/16 (192.168.0.0 - 192.168.255.255)

Special IP Addresses

- **Loopback**: 127.0.0.1 (localhost)
- **APIPA**: 169.254.0.0/16 (Automatic Private IP)
- **Broadcast**: 255.255.255.255
- **Default Route**: 0.0.0.0/0

Subnetting & CIDR

Subnetting Process

1. Determine number of required subnets/hosts
2. Calculate subnet bits needed
3. Create new subnet mask
4. Calculate subnet ranges

CIDR (Classless Inter-Domain Routing)

- **Format**: IP/prefix (e.g., 192.168.1.0/24)
- **Benefits**: Efficient IP allocation, route aggregation

- **VLSM**: Variable Length Subnet Masking

Subnetting Example

```
Network: 192.168.1.0/24
Requirement: 4 subnets

Solution:
- Borrow 2 bits from host portion
- New subnet mask: /26 (255.255.255.192)
- Subnets:
  1. 192.168.1.0/26    (1-62)
  2. 192.168.1.64/26   (65-126)
  3. 192.168.1.128/26  (129-190)
  4. 192.168.1.192/26  (193-254)
```

IPv6 Addressing

IPv6 Features

- **Address Length**: 128 bits

- **Format**: 8 groups of 4 hexadecimal digits
- **Example**: 2001:0db8:85a3:0000:0000:8a2e:0370:7334
- **Simplified**: 2001:db8:85a3::8a2e:370:7334

IPv6 Address Types

- **Unicast**: One-to-one communication
- **Multicast**: One-to-many communication
- **Anycast**: One-to-nearest communication

NAT (Network Address Translation)

NAT Types

- **Static NAT**: One-to-one mapping
- **Dynamic NAT**: Pool of public IPs
- **PAT (Port Address Translation)**: Many-to-one with port mapping

NAT Benefits & Drawbacks

Benefits:

- IP address conservation
- Security through obscurity
- Network isolation

Drawbacks:

- Breaks end-to-end connectivity
- Complicates certain protocols
- Performance overhead

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4. Protocols & Ports

Protocol Categories

Application Layer Protocols

Protocol	Port	Purpose	Security Considerations
HTTP	80	Web traffic	Unencrypted, vulnerable to interception

Protocol	Port	Purpose	Security Considerations
HTTPS	443	Secure web traffic	Encrypted with SSL/TLS
FTP	20/21	File transfer	Plain text, use SFTP instead
SFTP	22	Secure file transfer	Encrypted file transfer
SSH	22	Secure remote access	Encrypted remote shell
Telnet	23	Remote terminal	Plain text, avoid usage
SMTP	25	Email sending	Can be relayed, use authentication
DNS	53	Domain name resolution	Vulnerable to poisoning attacks
DHCP	67/68	Dynamic IP assignment	Rogue DHCP server attacks
POP3	110	Email retrieval	Plain text, use secure versions
IMAP	143	Email access	Better than POP3, use IMAPS
SNMP	161	Network management	SNMPv1/v2 insecure, use v3
LDAP	389	Directory services	Use LDAPS (636) for security
SMB	445	File/printer sharing	Vulnerable to various attacks

Protocol	Port	Purpose	Security Considerations
RDP	3389	Remote desktop	Enable NLA, use VPN

Transport Layer Protocols

Protocol	Features	Use Cases	Security Notes
TCP	Reliable, connection-oriented	Web, email, file transfer	Vulnerable to hijacking
UDP	Fast, connectionless	DNS, DHCP, streaming	No built-in security

Network Layer Protocols

Protocol	Purpose	Security Relevance
IP	Packet routing	Source routing attacks
ICMP	Error reporting	Ping of death, reconnaissance
ARP	MAC address resolution	ARP spoofing attacks
OSPF	Dynamic routing	Routing table manipulation
BGP	Internet routing	BGP hijacking

Port Memorization Tricks

Common Port Mnemonics

- **HTTP (80)**: "HTTP ate-y" (80)
- **HTTPS (443)**: "HTTPS for-for-three" (443)
- **SSH (22)**: "SSH two-two" (22)
- **FTP (21)**: "FTP twenty-one" (21)
- **SMTP (25)**: "SMTP twenty-five" (25)
- **DNS (53)**: "DNS fifty-three" (53)

Port Ranges

- **Well-known**: 0-1023 (System/privileged)
- **Registered**: 1024-49151 (User/application)
- **Dynamic**: 49152-65535 (Private/temporary)

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5. Network Topologies

Physical Topologies

Bus Topology

- **Structure**: Single cable backbone
- **Advantages**: Simple, cost-effective
- **Disadvantages**: Single point of failure, collision domain
- **Security**: Easy to tap into cable

Star Topology

- **Structure**: Central hub/switch with spokes to devices
- **Advantages**: Easy troubleshooting, no single cable failure
- **Disadvantages**: Central device failure affects all
- **Security**: Central monitoring point

Mesh Topology

- **Full Mesh**: Every device connected to every other
- **Partial Mesh**: Some devices have multiple connections

- **Advantages**: High redundancy, fault tolerance
- **Disadvantages**: Complex, expensive
- **Security**: Multiple paths complicate monitoring

Hybrid Topology

- **Structure**: Combination of multiple topologies
- **Example**: Star-Bus, Star-Ring
- **Advantages**: Scalable, flexible
- **Security**: Requires comprehensive security strategy

Wireless Topologies

Infrastructure Mode

- **Structure**: Devices connect through Access Points
- **Security**: WPA2/WPA3 encryption, RADIUS authentication

Ad-hoc Mode

- **Structure**: Device-to-device connections
- **Security**: Difficult to secure, avoid in production

Mesh Networks

- **Structure**: Wireless devices form interconnected network
- **Security**: Encryption at multiple levels

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6. Routing & Switching

Routing Concepts

Static vs Dynamic Routing

Aspect	Static Routing	Dynamic Routing
Configuration	Manual route entries	Automatic route discovery
Scalability	Poor for large networks	Good scalability

Aspect	Static Routing	Dynamic Routing
Bandwidth Usage	No routing overhead	Uses bandwidth for updates
Convergence	No convergence time	Takes time to converge
Security	More secure (no routing updates)	Vulnerable to routing attacks

Dynamic Routing Protocols

Distance Vector Protocols

- **RIP (Routing Information Protocol)**:
  - **Metric**: Hop count (max 15)
  - **Updates**: Every 30 seconds
  - **Vulnerabilities**: Slow convergence, routing loops

Link-State Protocols

- **OSPF (Open Shortest Path First)**:
  - **Metric**: Cost (based on bandwidth)

- **Features**: Fast convergence, hierarchical design
- **Security**: Authentication support

Path Vector Protocols

- **BGP (Border Gateway Protocol)**:
  - **Use**: Internet routing between AS
  - **Vulnerabilities**: BGP hijacking, route leaks

Switching Concepts

MAC Address Learning

1. Switch receives frame
2. Records source MAC in address table
3. Forwards frame based on destination MAC
4. Floods if destination unknown

VLAN (Virtual LAN)

- **Purpose**: Logical network segmentation

- **Benefits**:
  - Broadcast domain isolation
  - Security through segmentation
  - Flexible network design
- **Types**: Port-based, MAC-based, Protocol-based

VLAN Configuration Example

```
VLAN 10: Accounting Department
VLAN 20: IT Department
VLAN 30: Guest Network
VLAN 99: Management VLAN
```

Trunking

- **Purpose**: Carry multiple VLANs over single link
- **Protocols**:
  - **802.1Q**: Standard trunking protocol
  - **ISL**: Cisco proprietary (deprecated)

- **Security**: VLAN hopping attacks possible

STP (Spanning Tree Protocol)

- **Purpose**: Prevent switching loops
- **Process**:
  1. Elect root bridge
  2. Calculate shortest paths
  3. Block redundant paths
- **Security**: Root bridge manipulation attacks

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7. Firewalls & VPNs

Firewall Types & Technologies

Packet Filtering Firewalls

- **Operation**: Examine individual packets
- **Criteria**: Source/destination IP, ports, protocols

- **Limitations**: No state tracking, application awareness

Stateful Inspection Firewalls

- **Operation**: Track connection states
- **Benefits**: Better security than packet filtering
- **Features**: Connection table maintenance

Application Layer Firewalls

- **Operation**: Deep packet inspection
- **Benefits**: Application-specific rules
- **Examples**: Web Application Firewalls (WAF)

Next-Generation Firewalls (NGFW)

- **Features**:
  - Application awareness
  - Intrusion prevention
  - User identity integration

- Threat intelligence

Firewall Rules & Policies

Default Rule Principles

- **Deny by Default**: Block all traffic except explicitly allowed
- **Least Privilege**: Minimum necessary access
- **Rule Order**: Most specific to least specific

Example Firewall Rules

```
1. ALLOW: TCP 192.168.1.0/24 → ANY:80,443 (Web browsing)
2. ALLOW: TCP 192.168.1.100 → 10.0.0.5:22 (Admin SSH)
3. DENY: ANY → 192.168.1.0/24:22 (Block external SSH)
4. DENY: ALL (Default deny)
```

VPN Technologies

Site-to-Site VPN

- **Purpose**: Connect entire networks
- **Protocols**: IPSec, GRE
- **Use Cases**: Branch office connectivity

Remote Access VPN

- **Purpose**: Individual user access
- **Protocols**: SSL/TLS, IPSec
- **Use Cases**: Remote workers, BYOD

VPN Protocols Comparison

Protocol	Layer	Security	Performance	Use Case
IPSec	Network	High	Good	Site-to-site
SSL/TLS	Application	High	Moderate	Remote access
PPTP	Data Link	Low	High	Legacy (avoid)
L2TP/IPSec	Data Link	High	Moderate	Remote access

VPN Security Considerations

- **Encryption Strength**: Use AES-256
- **Authentication**: Multi-factor preferred
- **Key Management**: Proper key rotation
- **Split Tunneling**: Security vs performance trade-off

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8. Network Attacks

Reconnaissance Attacks

Network Scanning

- **Ping Sweeps**: Discover live hosts
- **Port Scans**: Identify open services
- **Service Enumeration**: Gather service details
- **OS Fingerprinting**: Identify operating systems

DNS Reconnaissance

- **Zone Transfers**: Download DNS records

- **DNS Brute Force**: Discover subdomains
- **Reverse DNS**: IP to hostname mapping

Man-in-the-Middle (MITM) Attacks

ARP Spoofing

- **Method**: Send fake ARP responses
- **Result**: Redirect traffic through attacker
- **Tools**: Ettercap, Bettercap
- **Prevention**: Static ARP entries, ARP inspection

DNS Spoofing/Poisoning

- **Method**: Provide false DNS responses
- **Result**: Redirect users to malicious sites
- **Prevention**: DNS over HTTPS (DoH), DNSSEC

SSL Stripping

- **Method**: Downgrade HTTPS to HTTP
- **Result**: Intercept credentials
- **Prevention**: HSTS headers, certificate pinning

Network Layer Attacks

IP Spoofing

- **Method**: Forge source IP addresses
- **Use Cases**: DoS amplification, bypassing filters
- **Prevention**: Ingress/egress filtering

ICMP Attacks

- **Ping of Death**: Oversized ICMP packets
- **ICMP Floods**: Overwhelm target with pings
- **Prevention**: ICMP rate limiting, filtering

Transport Layer Attacks

TCP Hijacking

- **Method**: Predict sequence numbers
- **Result**: Take over established connections
- **Prevention**: Random sequence numbers, encryption

SYN Flood

- **Method**: Send many SYN packets without completing handshake
- **Result**: Exhaust connection resources
- **Prevention**: SYN cookies, rate limiting

Application Layer Attacks

HTTP-based Attacks

- **HTTP Flooding**: Overwhelm web servers
- **Slowloris**: Keep connections open with slow requests
- **Prevention**: Rate limiting, timeout configurations

Denial of Service (DoS) & Distributed DoS (DDoS)

DoS Attack Types

- **Volume-based**: Bandwidth exhaustion
- **Protocol-based**: Resource exhaustion
- **Application-based**: Application resource exhaustion

DDoS Amplification

- **DNS Amplification**: Small queries → Large responses
- **NTP Amplification**: Exploit NTP monlist command
- **Prevention**: Disable amplification services, rate limiting

Attack Scenarios & Examples

Internal Network Compromise

1. Initial access via phishing/malware
2. Network reconnaissance (Nmap, Nessus)

3. Lateral movement (credential harvesting)
4. Privilege escalation
5. Data exfiltration

Network Perimeter Breach

1. External reconnaissance
2. Vulnerability exploitation
3. Firewall/IPS evasion
4. Internal network access
5. Persistence establishment

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9. Network Pentesting Tools

Network Discovery & Scanning

Nmap (Network Mapper)

- **Purpose**: Network discovery and security auditing

- **Key Features**:
  - Host discovery
  - Port scanning
  - Service detection
  - OS fingerprinting
  - Vulnerability scanning (NSE scripts)

Common Nmap Commands:

```
# Basic host discovery
nmap -sn 192.168.1.0/24

# TCP SYN scan (stealth)
nmap -sS target.com

# Service version detection
nmap -sV target.com

# OS detection
nmap -O target.com
```

```
# Aggressive scan with scripts
nmap -A target.com

# Specific ports
nmap -p 80,443,22 target.com

# UDP scan
nmap -sU target.com
```

Netcat (The Swiss Army Knife)

- **Purpose**: Network communication utility
- **Use Cases**:
  - Port scanning
  - Banner grabbing
  - File transfers
  - Backdoor creation (ethical testing)

Netcat Examples:

```
# Port scanning
nc -zv target.com 1-1000

# Banner grabbing
nc target.com 80

# Listen for connections
nc -l -p 4444

# File transfer
nc -l -p 1234 > received_file.txt
nc target_ip 1234 < file_to_send.txt
```

Traffic Analysis & Sniffing

Wireshark

- **Purpose**: Network protocol analyzer
- **Key Features**:
  - Live packet capture

- Deep protocol inspection
- Filtering capabilities
- Statistical analysis

Wireshark Filters:

```
# HTTP traffic
http

# Specific IP
ip.addr == 192.168.1.100

# TCP traffic on port 80
tcp.port == 80

# Failed TCP connections
tcp.flags.reset == 1

# DNS queries
dns.qry.name contains "example.com"
```

Tcpdump

- **Purpose**: Command-line packet analyzer
- **Benefits**: Lightweight, scriptable

Tcpdump Examples:

```
# Capture all traffic
tcpdump -i eth0

# Capture HTTP traffic
tcpdump -i eth0 port 80

# Save to file
tcpdump -i eth0 -w capture.pcap

# Read from file
tcpdump -r capture.pcap
```

```
# Capture with verbose output
tcpdump -i eth0 -v host 192.168.1.100
```

Network Attack Tools

Ettercap

- **Purpose**: MITM attacks and network sniffing
- **Capabilities**:
  - ARP spoofing
  - DNS spoofing
  - Password collection
  - Traffic manipulation

Hping3

- **Purpose**: Network testing and attack tool
- **Use Cases**:
  - Custom packet crafting
  - Firewall testing

- DoS testing (ethical)

Hping3 Examples:

```
# SYN flood test
hping3 -S --flood -V target.com

# Ping with custom packet size
hping3 -1 -d 1000 target.com

# Traceroute with TCP
hping3 -T -S -p 80 target.com
```

Web Application Testing

Burp Suite

- **Purpose**: Web application security testing
- **Key Features**:
  - HTTP/HTTPS proxy

    - Request/response modification
    - Automated scanning
    - Intruder (brute force)

OWASP ZAP

- **Purpose**: Web application vulnerability scanner
- **Benefits**: Free, user-friendly
- **Features**: Automated scanning, manual testing tools

Network Vulnerability Scanners

Nessus

- **Purpose**: Comprehensive vulnerability scanning
- **Features**:
    - Plugin-based scanning
    - Compliance checking
    - Reporting capabilities

OpenVAS

- **Purpose**: Open-source vulnerability scanner
- **Benefits**: Free, comprehensive
- **Components**: Scanner, manager, client interface

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10. Real-World Cybersecurity Scenarios

Scenario 1: Corporate Network Breach

Attack Timeline

1. **Reconnaissance Phase**
   - External port scans reveal open RDP (3389)
   - Social engineering gathers employee information
   - DNS enumeration discovers internal hostnames
2. **Initial Compromise**
   - Brute force attack on RDP succeeds

- Malware deployment establishes persistence
- Local privilege escalation

3. **Lateral Movement**
    - Network scanning identifies domain controllers
    - Credential harvesting from compromised systems
    - Pass-the-hash attacks
4. **Data Exfiltration**
    - Database access through compromised accounts
    - Encrypted tunnels to avoid detection
    - Staged data extraction over time

Defense Strategies

- **Network Segmentation**: Limit lateral movement
- **Multi-factor Authentication**: Prevent credential-based access
- **Monitoring**: SIEM for anomaly detection
- **Endpoint Protection**: Prevent malware execution
- **Regular Patching**: Close vulnerability windows

Scenario 2: DDoS Attack on E-commerce Site

Attack Pattern

1. **Reconnaissance**
   - Identify application bottlenecks
   - Test rate limiting mechanisms
   - Map content delivery network
2. **Multi-vector Attack**
   - Volumetric attack (bandwidth exhaustion)
   - Protocol attack (state table exhaustion)
   - Application layer attack (resource exhaustion)
3. **Evasion Techniques**
   - IP address rotation
   - Legitimate-looking traffic patterns
   - Distributed attack sources

Mitigation Strategies

- **DDoS Protection Services**: Cloud-based filtering
- **Rate Limiting**: Application and network level
- **Failover Systems**: Maintain availability
- **Incident Response**: Rapid threat mitigation

Scenario 3: Insider Threat Network Activity

Threat Indicators

- **Data Access Patterns**: Unusual file access
- **Network Traffic**: Data transfers to external sites
- **System Behavior**: Privilege escalation attempts
- **Timeline Analysis**: After-hours activity

Detection Methods

- **User Behavior Analytics**: Baseline normal behavior
- **Network Monitoring**: Traffic analysis
- **Data Loss Prevention**: Content inspection

- **Access Logging**: Comprehensive audit trails

Scenario 4: IoT Botnet Formation

Attack Progression

1. **Device Discovery**: Internet-wide scanning
2. **Vulnerability Exploitation**: Default credentials
3. **Malware Installation**: Persistent access
4. **Command and Control**: Centralized management
5. **Botnet Utilization**: DDoS, cryptocurrency mining

Defense Approach

- **Device Management**: Regular updates, credential changes
- **Network Monitoring**: Unusual traffic patterns
- **Segmentation**: Isolate IoT devices
- **Threat Intelligence**: Known malware signatures

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11. Quick Revision & Practice

Key Concepts Checklist

Networking Fundamentals ✓

- [ ] OSI vs TCP/IP model layers
- [ ] Network types (LAN, WAN, MAN)
- [ ] Private vs public IP ranges
- [ ] Subnet mask calculations

Network Devices ✓

- [ ] Router vs Switch vs Hub functions
- [ ] Firewall types and placement
- [ ] IDS vs IPS capabilities
- [ ] Proxy server configurations

Protocols & Ports ✓

- [ ] Common port numbers (HTTP, HTTPS, SSH, FTP, etc.)
- [ ] TCP vs UDP characteristics
- [ ] DNS, DHCP, ARP operations
- [ ] Secure vs insecure protocol alternatives

Network Attacks ✓

- [ ] MITM attack methods
- [ ] DoS/DDoS attack types
- [ ] ARP spoofing mechanics
- [ ] DNS poisoning techniques

Security Tools ✓

- [ ] Nmap scanning techniques
- [ ] Wireshark filtering
- [ ] Network monitoring approaches
- [ ] Vulnerability assessment methods

Multiple Choice Questions (MCQs)

Question 1

Which OSI layer is responsible for routing packets between networks? A) Transport Layer B) Network Layer ✓ C) Data Link Layer D) Session Layer

Question 2

What is the default port for HTTPS? A) 80 B) 22 C) 443 ✓ D) 25

Question 3

Which attack involves sending fake ARP responses to redirect traffic? A) DNS Spoofing B) ARP Spoofing ✓ C) IP Spoofing D) MAC Flooding

Question 4

What does NAT stand for? A) Network Access Translation B) Network Address Translation ✓ C) Network Application Transfer D) Network Authentication Token

Question 5

Which tool is best for deep packet inspection and protocol analysis? A) Nmap B) Netcat C) Wireshark ✓ D) Hping3

Practical Lab Ideas

Lab 1: Network Discovery

Objective: Discover live hosts and services on a network **Tools**: Nmap, ping **Steps**:

1. Perform ping sweep: `nmap -sn 192.168.1.0/24`
2. Port scan discovered hosts: `nmap -sS -p 1-1000 [target]`
3. Service version detection: `nmap -sV [target]`
4. Document findings and analyze results

Lab 2: Traffic Analysis

Objective: Capture and analyze network traffic **Tools**: Wireshark, tcpdump **Steps**:

1. Set up packet capture on network interface
2. Generate different types of traffic (HTTP, DNS, FTP)
3. Apply filters to isolate specific protocols
4. Analyze packet contents and identify potential issues

Lab 3: Firewall Rule Testing

Objective: Configure and test firewall rules

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📢 Telegram Channel (Updates & Resources):
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🗣️ Discord Server (Voice & Community Chat):
https://discord.gg/qA3CqZ86hY
👻 Snapchat (Cyber Moments):
https://www.snapchat.com/add/cybermindspace?share_id=8WsnFshDsWg&locale=en-IN
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https://x.com/cybermindspace?s=21

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